A Novel Scene-Aware Pipeline Architecture for Spaceborne SAR Regional Imaging and Ship Target Detection

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***Abstract*—With the rapid advancement of synthetic aperture radar (SAR) technology and the explosive growth of on-orbit data, conventional processing frameworks are increasingly constrained by high resource consumption and substantial latency, rendering them inadequate for real-time maritime ship detection on resource-limited platforms. To address this challenge, this article proposes a novel scene-aware imaging-detection integration (SAIDI) architecture for stripmap-mode SAR, which unifies traditionally decoupled imaging and detection tasks into a cascaded processing pipeline. Specifically, the framework comprises four key components: an imaging necessity judgment module to filter target-free raw echoes, a fast imaging algorithm bypassing range cell migration correction, a lightweight pruned YOLOv11 detection network, and a dedicated ship dataset constructed from real satellite measurements. Experiments on real satellite data demonstrate that the proposed method maintains robust detection performance in both open-ocean and coastal scenarios while eliminating redundant computations in background regions. Compared to conventional methods, the proposed scheme significantly reduces processing latency and peak memory footprint, while optimizing CPU utilization and power efficiency. In addition, this method is mainly designed for medium- and large-sized marine ships.**

***Index Terms*—Imaging algorithm, pipeline-based architecture for imaging and detection, synthetic aperture radar (SAR), target detection.**

I. Introduction

SYNTHETIC aperture radar (SAR) has become a core technical means for marine ship target detection owing to its all-day, all-weather observation performance and cloud penetration ability [1]. It covers extensive application scenarios, such as open-sea traffic monitoring, coastal security and defense, as well as emergency response to marine disasters [2]. Recent advances in spaceborne SAR technologies have greatly aggravated the massive data throughput bottleneck, which substantially raises the practical demand for real-time ship detection. Against this backdrop, on-board real-time ship detection has become an urgent technical requirement for spaceborne SAR platforms to support timely maritime situational awareness [3], [4]. Nevertheless, on-orbit data processing is subject to inherent hardware limitations of satellite platforms, including insufficient computing power and storage space, thereby placing strict constraints on the operational efficiency and lightweight design of detection algorithms [5], [6].

Conventional SAR data processing follows the sequential "imaging-first, detection-later" pipeline, where imaging results act as the basic input for subsequent ship detection [7]. Accordingly, existing SAR ship detection studies have long optimized imaging and detection modules separately. The imaging module targets high-fidelity imaging via resolution enhancement and sea clutter suppression [8], [9], [10]. The detection module aims for superior accuracy and inference speed, supported by two mainstream technical routes: classical detectors [11], [12], [13], [14] and deep learning frameworks [15], [16], [17], [18], [19], [20], including applications in infrared remote sensing [21], [22], [23]. Although this separate optimization scheme achieves satisfactory performance in ground offline processing, it is incompatible with the stringent hardware and latency limitations of on-board real-time SAR processing [24].

Against the aforementioned real-time processing constraints, two distinct acceleration strategies have been developed for the SAR imaging-detection pipeline. As the primary solution, imaging and detection submodules are accelerated independently. For imaging optimization, computational overhead can be cut via two paths. Software optimizations include adaptive data compression [25] and sparse sampling [26] to reduce reconstruction cost, simplified compensation terms [27], [28], [29], and relaxed interpolation precision [30], [31]. Alternatively, hardware parallel acceleration is implemented

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on CPUs [32], GPUs [33], DSPs [34], and FPGAs [35]. For detection optimization, multidimensional lightweight designs are widely adopted. Model-level optimizations adopt network pruning [36], knowledge distillation [37], and low-rank decomposition [38] to cut redundant parameters and computation; feature-level optimizations deploy lightweight backbones to streamline feature extraction [39], [40], [17]; input-level optimizations narrow valid data proposal algorithms [41]. While these methods boost local efficiency to some degree, they retain the conventional serial pipeline: echo acquisition → full-scene imaging → full-image detection [42], [43]. Every subsequent processing stage can only start after the prior stage is fully finished. Accordingly, these local optimizations can only cut a small portion of total computation load, resulting in limited overall acceleration gains for on-board real-time tasks.

Unlike the separate optimization paradigm, the second approach leverages inherent features of raw echoes and range-compressed data to jointly accelerate the complete imaging-detection pipeline. Driven by this integrated processing trend, raw-echo and range-compressed domains have become two mainstream research branches in recent years [44], [45], it can directly accomplish target screening in the transform domain and improve processing efficiency by reducing redundant computations [46], [47], [48]. Methods based on raw echoes identify ship targets directly according to the inherent scattering characteristics of ships within unprocessed echo data [49], [50]. In contrast, algorithms operating in the range-compressed domain execute coarse target screening after range compression; reconstruction and refined secondary detection are merely carried out over regions with suspected ship signatures [51], [45], [52], [53]. Both types of integrated strategies center on ship detection as the core objective, thus avoiding full-scene reconstruction and drastically reducing redundant computational overhead. This integrated processing scheme achieves favorable performance for open-ocean scenarios with sparse ship targets. Nevertheless, its generalization capability across diverse scenes remains constrained, and it has to fall back to independent module acceleration when confronting complex nearshore environments.

Beyond their respective application defects, the two acceleration strategies also feature inherent limitations for on-board real-time processing. These methods require full raw echo acquisition before computation, introducing unavoidable processing latency and imposing full-scene-level memory overhead. Consequently, computational flow simplification only delivers marginal acceleration and cannot mitigate the above constraints. Among mainstream SAR operating modes, stripmap mode is distinguished by a fixed antenna beam pointing configuration. Under this geometric setup, targets within the imaging swath receive uniform illumination while traversing the beam at a constant platform velocity. Unlike sliding spotlight or TOPS modes, this stable scanning mechanism facilitates long-duration continuous imaging and ground target detection. Consequently, this mechanism can produce echo signals with a substantially larger azimuthal extent than the range dimension. This inherent feature of the stripmap mode, combined with the requirement for full echo collection before computation shared by all aforementioned methods, significantly exacerbates the memory and latency constraints discussed above. Both memory overhead and processing latency increase linearly with azimuth aperture, worsening the inherent resource limitations for extended-scene imaging. Such severe constraints substantially degrade the on-board real-time processing capacity. Nevertheless, existing research fails to address this critical bottleneck.

To tackle the aforementioned limitations, this article proposes a scene-aware imaging-detection integration (SAIDI) method for stripmap SAR. Unlike conventional sequential workflows that separate imaging and target detection, the proposed method adopts a cascaded processing collaborative processing paradigm for joint computational acceleration. A strip-by-strip progressive processing strategy removes the need to buffer full raw echoes and drastically cuts on-board memory usage. In addition, considering the divergent echo characteristics of sparse open-ocean ships and complex inshore targets, SAIDI evaluates each echo segment via statistical features before imaging and discards target-free segments to reduce redundant computation. Extensive experiments on real spaceborne SAR data demonstrate that our method reduces latency, CPU load, and memory footprint without sacrificing detection accuracy, and it fits well with on-board real-time detection of medium and large ships for both open sea and inshore complicated scenes.

The rest of this article is organized as follows. Section II covers related work and the dataset used in this study. Section III presents the proposed scene-aware pipeline architecture. Experimental results and corresponding analysis are provided in Section IV. Section V concludes this article.

II. Related Work and Dataset

A. Segmentation Method in Azimuth

In the proposed architecture, raw echo data are partitioned along the azimuth dimension to enhance processing efficiency and reduce peak memory consumption. The partitioning strategy is formulated based on three inherent constraints.

The first design constraint concerns the SAR imaging principle. Azimuth samples with incomplete synthetic aperture integration must be truncated after image formation. The number of truncated samples is determined by the synthetic aperture time, which is calculated from the system imaging parameters, as given by

$$T_a = \lambda \cdot R_ref / V_g / L_a \tag{1}$$

where λ denotes the carrier wavelength, R_ref denotes the reference slant range, V_g denotes the platform ground velocity, and L_a denotes the physical length of the antenna [54].

The number of azimuth synthetic aperture accumulation points can be further derived from the synthetic aperture time T_s, as formulated in the following equation:

$$N_{T_a} = PRF \cdot T_a \tag{2}$$

where PRF denotes the pulse repetition frequency.

Accordingly, the sample count of each azimuth block must be greater than the number of azimuth synthetic aperture

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integration samples during azimuth echo partitioning. Furthermore, to ensure the continuity of the imaged scene, the minimum number of overlapping samples between adjacent blocks in the original azimuth echo is equal to the number of synthetic aperture points, as illustrated in Fig. 1.

The second design constraint relates to the integrity of imaged ship targets, which prevents targets from being split during azimuth echo partitioning so as to avoid degradation in detection performance. Accordingly, an overlapping region is reserved between adjacent imaging blocks in the segmentation process. The size of this overlapping region is calculated based on the typical physical dimensions of ship targets, as given by

$$N_{ship_a} = \frac{h_{ship}}{n_a} = \frac{h_{ship} \cdot PRF}{V_g} \tag{3}$$

$$N_{ship_r} = \frac{h_{ship}}{n_r} = \frac{h_{ship} \cdot 2f_s}{c} \tag{4}$$

where N_{ship_a} and N_{ship_r} denote the number of pixels corresponding to the ship target size in the azimuth and range dimensions, respectively; h_{ship} represents the typical physical size of the ship target; n_a and n_r are the azimuth and range pixel spacing, respectively; f_s is the signal sampling rate; and c denotes the speed of light.

This requirement imposes a constraint on the size of each azimuth block: the number of azimuth points per block must be greater than twice the width of the overlapping region between adjacent imaging results, such that the remaining nonoverlapping portion can be cropped to a complete target-containing region, which can be directly fed into the target detection network.

The third design constraint involves computational efficiency, which is quantitatively formulated in the following equation:

$$N_a = N_{T_a} + 2 \cdot N_{ship_a} + N_{cut_a}. \tag{5}$$

To accelerate the Fourier transform operations in the imaging procedure, the radix-2 butterfly FFT algorithm is adopted. Accordingly, range samples are zero-padded to the nearest integer power of two, while azimuth blocks are partitioned such that the number of azimuth pulses per block is set to an exact power of two, eliminating the additional overhead of azimuth-direction zero padding.

In summary, the criterion for determining the azimuth sample count per block is summarized in the following equation to satisfy all aforementioned constraints:

$$\min\{2^n \mid n \in Z, 2^n > N_a\} = 2^{\lceil \log_2 N_a \rceil}. \tag{6}$$

B. No-RCMC Imaging Algorithm

In this work, target detection is taken as the sole core objective; thus, nonessential imaging accuracy can be traded off for improved computational efficiency.

For stripmap-mode SAR imaging, the spatial resolution is generally moderate, and the impact of range migration on imaging results is relatively limited. In addition, the significant difference in backscattering coefficients between ship targets and the sea-surface background provides a solid discriminative basis for target detection, making the detection task robust to moderate imaging defocus. Therefore, for sea-surface ship detection tasks that do not require high-resolution imaging capabilities such as target recognition, omitting range cell migration correction (RCMC) will not exert a substantial impact on detection performance.

Specifically, such a precision-for-efficiency tradeoff is feasible as long as the weakened focusing effect does not cause the loss of key target features, such as ship contours and strong scattering points. To further verify the feasibility of this simplification, this section quantifies the impact of range migration on imaging results under the current scenario through mathematical derivation. The expression of the raw echo signal is given in the following equation:

$$\begin{aligned} S_{echo}(\tau, t; r) = {} & \sigma \cdot \omega_a(\eta - \eta_c) \cdot \omega_r\left[\tau - 2R(\eta)/c\right] \\ & \cdot \exp\left\{-j4\pi f_0 \frac{R(\eta)}{c}\right\} \\ & \cdot \exp\left\{j\pi k_r \left[\tau - \frac{2R(\eta)}{c}\right]^2\right\} \end{aligned} \tag{7}$$

where σ denotes the backscattering coefficient of the illuminated target, τ is the range time, η is the azimuth time, η_c is the beam center deviation time, $\omega_r(\tau)$ is the range envelope, $\omega_a(\tau)$ is the azimuth envelope, f_0 is the radar center frequency, K_r is the range chirp rate, and $R(\eta)$ is the instantaneous slant range.

The instantaneous slant range $R(\eta)$ of the target can be expanded into a Taylor series with respect to azimuth time ta, as formulated in the following equation:

$$R(\eta) = \sqrt{R_0^2 + V_r^2\eta^2} \approx R_0 + \frac{V_r^2\eta^2}{2R_0} \tag{8}$$

where R_0 denotes the closest slant range and V_r denotes the equivalent radar velocity.

When only pulse compression is applied in both range and azimuth dimensions, the corresponding range and azimuth compensation factors are provided in the following equations:

$$H_{ra}(f_\tau) = rect\left\{\frac{f}{|K|T}\right\} \cdot \exp\left\{j\pi\frac{f^2}{K}\right\} \tag{9}$$

$$H_{az}(f_\eta) = \exp\left\{-j\pi\frac{f_\eta^2}{K_a}\right\}. \tag{10}$$

Following the matched filtering operation, the final imaging result is formulated as

$$\begin{aligned} S_{AC}(\tau, \eta) = {} & \sigma \cdot p_r\left[\tau - 2R(\eta)/c\right] p_a(\eta) \\ & \cdot \exp\left\{-j\frac{4\pi f_0 R_0}{c}\right\} \exp\left\{j2\pi f_{\eta_c}\eta\right\}. \end{aligned} \tag{11}$$

For comparative analysis, when precise RMC is implemented via interpolation, the fully processed ideal point target signal can be expressed as follows:

$$\begin{aligned} S_{right}(\tau, \eta) = {} & \sigma \cdot p_r\left[\tau - 2R_0/c\right] p_a(\eta) \\ & \cdot \exp\left\{-j\frac{4\pi f_0 R_0}{c}\right\} \exp\left\{j2\pi f_{\eta_c}\eta\right\}. \end{aligned} \tag{12}$$

The derivation shows that the core difference between the cases with and without RCMC lies in the range envelope term:

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TABLE I
SIMULATION AND CALCULATION PARAMETERS BASED ON THE REAL SATELLITE

Parameters	Value	Unit
Equivalent Radar Speed	7000	m/s
Synthetic Aperture Time	0.6	s
Closest Slant Range	850	km
Signal Bandwidth	85	MHz

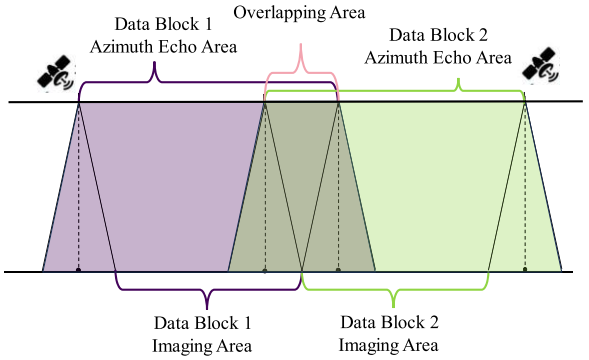


Fig. 1. Stripmap-mode SAR imaging schematic.

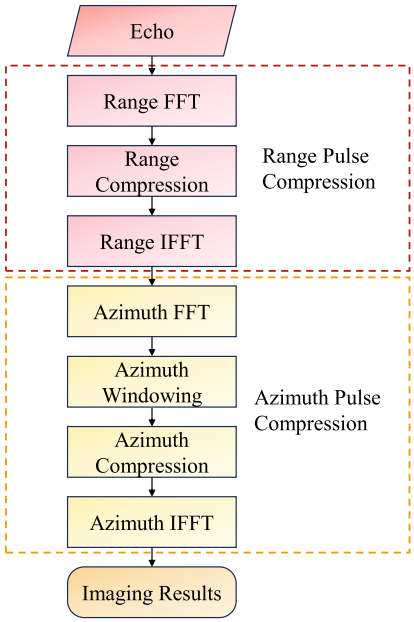


Fig. 2. Workflow of the No-RCMC algorithm.

the scheme with RCMC corrects the envelope shift varying with azimuth time, while the scheme without RCMC adopts the fixed range gate approximation. This discrepancy causes the echo energy of a point target to defocus across multiple range gates, and the peak position of the envelope shifts along the range dimension with azimuth time.

To quantitatively analyze the above effects, numerical simulations are performed based on the actual imaging parameters of an on-orbit satellite, with the specific system parameters listed in Table I.

Accordingly, the maximum slant range deviation induced by range migration is derived as follows:

$$\Delta R_{\max} = \frac{V_r^2 \eta_{\max}^2}{2R_0} = \frac{V_r^2 T_a^2}{8R_0} \approx 2.594\ m. \tag{13}$$

The corresponding range resolution is approximately

$$\rho_r = \frac{c}{2B_r} \approx 1.76. \tag{14}$$

Based on the above calculations, the number of range gates crossed by range migration is approximately

$$N = \left\lceil \frac{\Delta R_{\max}}{\rho_r} \right\rceil = 1. \tag{15}$$

Since this value is sufficiently small, range migration will not cause significant degradation in imaging focusing quality or subsequent detection performance, which is consistent with the previous qualitative analysis.

On this basis, the imaging algorithm adopted in this article only performs range and azimuth pulse compression to achieve echo focusing. This simplified imaging scheme is referred to as the No-RCMC algorithm throughout this article. Meanwhile, to suppress ghosting artifacts caused by azimuth ambiguity, azimuth windowing is implemented in the range–Doppler domain, preventing the detection network from misidentifying defocused ghosting artifacts as ship targets. The specific processing workflow is illustrated in Fig. 2.

C. Pruned Yolo for Ship Detection With Datasets

To ensure detection performance under the simplified No-RCMC imaging scheme and satisfy the real-time processing requirements of spaceborne platforms, two targeted adaptations are implemented on the detection side: construction of a matched training dataset, and lightweight optimization of the detection network via channel pruning.

For dataset construction, publicly available SAR ship detection datasets are not directly adopted, as omitting range migration correction leads to feature discrepancies between simplified imaging results and standard fully focused SAR images. Although such discrepancies exert no significant impact at the visual interpretation level, incompletely focused ship targets differ significantly from fully focused ones in key feature characteristics, including contour integrity and scattering intensity distribution. This may cause deviations in feature extraction by the detection network, further resulting in missed detections and false alarms in practical deployment, and ultimately degrading the cascaded processing performance of the integrated imaging-detection system. Accordingly, training data with consistent imaging conditions are essential to eliminate the feature mismatch between training and deployment and to guarantee the reliability of final detection results. To address this issue, the No-RCMC imaging algorithm is applied to process raw echo data of open-ocean scenarios acquired from on-orbit satellites. A dedicated ship detection dataset is constructed based on the simplified imaging results for subsequent detection model training. The dataset, denoted as the FAST_SHIP dataset, is detailed as follows in terms of data sources and composition.

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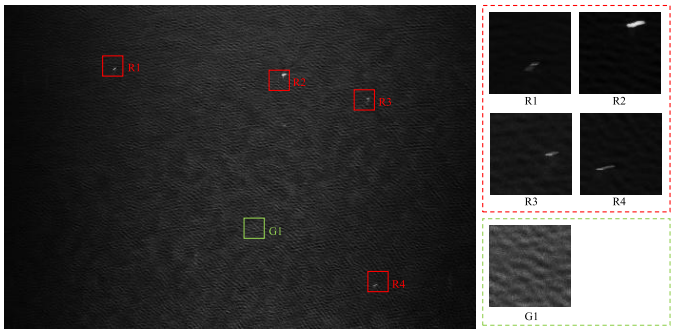


Fig. 3. FAST-SHIP dataset. Red bounding boxes mark positive ship samples and green bounding boxes mark negative samples.

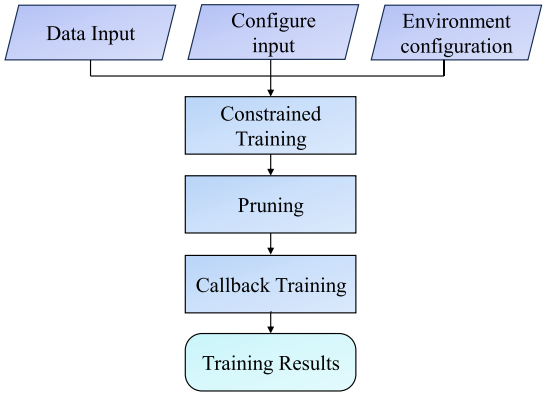


Fig. 4. Workflow description of the target detection module.

Specifically, 59 scenes of raw echo data from on-orbit satellites are processed via the proposed No-RCMC scheme. From cropped and quantized full-scene images, 1424 slices with ship targets are extracted, covering 1125 annotated ship instances and false targets. The FAST_SHIP dataset is randomly split 8:2 into training and validation sets for subsequent detection network training, with typical samples presented in Fig. 3.

For the lightweight optimization of the detection model, two complementary efficiency enhancement measures are adopted to pursue faster detection inference speed. Not only is the YOLOv11 model from the one-stage YOLO series selected as the baseline owing to its inherent high processing efficiency, but structured channel pruning is further implemented to eliminate redundant channel connections, so as to achieve additional acceleration while maintaining acceptable detection accuracy. The specific implementation workflow of the pruning and retraining procedure is illustrated in Fig. 4.

III. Scene-Aware Pipeline Architecture

This section systematically presents the echo-domain target prescreening mechanism, the intrablock processing pipeline of single-azimuth echoes, and the interblock parallel scheduling architecture of the overall system.

A. Target Existence Prediction Mechanism

Ship targets are sparsely distributed, especially in open-ocean scenarios [50]; thus, full processing of all echo data is unnecessary. To address this, a raw echo-domain target existence judgment module is integrated into the proposed architecture. This module skips the subsequent imaging procedure for data with no potential targets identified in the raw echo domain, so as to reduce wasted on-board processing time and hardware resources. The core identification principle of this mechanism is derived as follows.

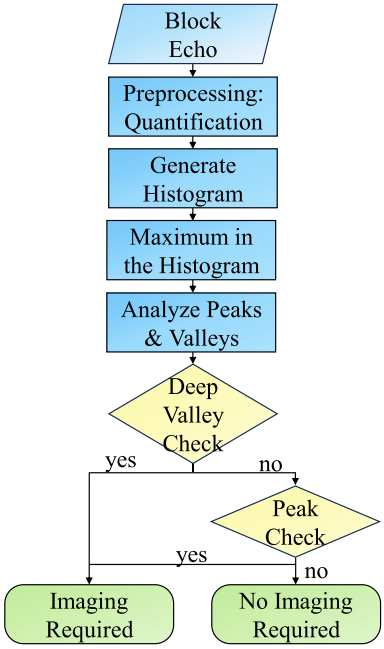


Fig. 5. Preimaging echo screening workflow.

In SAR imaging systems, the amplitude of transmitted signals remains constant. According to the radar equation, echo amplitude is primarily determined by the backscattering coefficients of targets in the observed scene. While also influenced by slant range, system parameters, and propagation loss, but the backscattering coefficient is the dominant factor driving echo amplitude differences. For ship detection, observation scenarios are primarily categorized into two main categories: open-ocean and harbor. Besides the marine background, typical scene components include ship targets, coastlines, man-made coastal structures, and false alarm sources. Diverse material properties result in backscattering coefficients spanning multiple orders of magnitude. Under most observation geometries, sea clutter echo amplitude is generally lower than that of other scene components. Notably, ship targets, primarily metallic with large radar cross sections (RCSs), have far stronger backscattering coefficients than the seawater background. This scattering characteristic constitutes the core physical basis for distinguishing ship targets from marine background in the raw echo domain. Leveraging this property, amplitude histogram analysis of raw echoes can be performed to quantify their statistical characteristics, supporting subsequent target presence judgment [55].

Based on the above physical principle, the workflow of the prescreening mechanism is illustrated in Fig. 5.

In data preprocessing, amplitude statistics are performed on all raw echo samples, and the amplitude range is adjusted via

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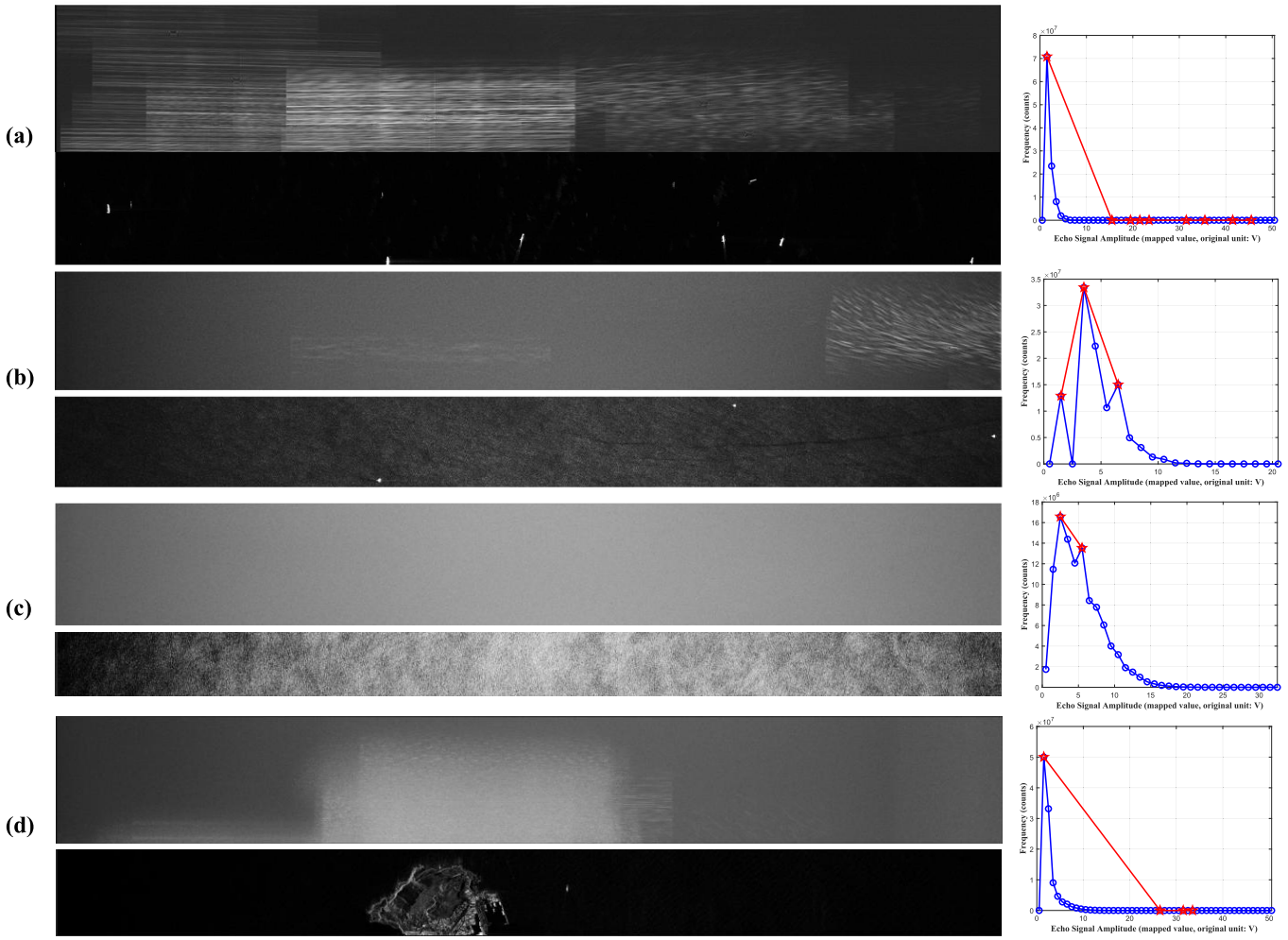


Fig. 6. Illustration of four typical echo scenarios: (a) standard ship scene, (b) ship scene with complex background, (c) pure ocean background scene, and (d) sea–land boundary scene. Each subfigure consists of three parts: the top-left panel shows the raw echo domain visualization; the bottom-left panel shows the amplitude image of the corresponding echo after full imaging processing; the right panel shows the amplitude histogram of the raw echo. In the histogram, the blue curve connects the statistical values of each amplitude bin, red asterisks mark local dominant peaks, and the red polyline connects these peaks to highlight the distribution characteristics.

mapping per the following equation:

$$\tilde{x}_i = \begin{cases} 20 \cdot \dfrac{x_i - X_{\min}}{X_{\max} - X_{\min}}, & X_{\max} < 30 \\ x_i, & 30 < X_{\max} < 50 \\ 50 \cdot \dfrac{x_i - X_{\min}}{X_{\max} - X_{\min}}, & X_{\max} > 50 \end{cases} \tag{16}$$

where x_i denotes the original amplitude before mapping, $X_{\min}$ and $X_{\max}$ denote the minimum and maximum original amplitudes, and $\tilde{x}_i$ denotes the mapped amplitude. The constant terms in the equation are derived from engineering experience.

An amplitude histogram is generated by traversing all samples with a unit step size. For intuitive illustration, four typical scenarios, including standard ship, complex-background ship, pure ocean, and sea–land boundary scenes, are compared in Fig. 6. Based on the histogram characteristics, target presence in the echo is comprehensively judged via three complementary criteria: the deep valley between adjacent dominant peaks, the peak intensity ratio, and the main peak position.

The first category addresses bimodal amplitude distributions where target echoes and sea clutter form two separable clusters. For strong-scattering targets, the large amplitude contrast creates a distinct interpeak valley; the valley depth ratio $R_{vp} = x_v/x_l$ is adopted to quantify cluster separation, where x_v is the valley amplitude and x_l is the amplitude of the peak on the left side of the valley.

For weak-scattering targets under rough sea conditions, the valley feature is indistinct but a secondary peak remains, so the adjacent peak amplitude ratio $R_{pp} = x_{\min}/x_{\max}$ serves as a supplementary metric, with $x_{\min}$ and $x_{\max}$ denoting the smaller and larger of the two adjacent peak amplitudes.

Analysis of real echo data shows the optimal R_{vp} separation threshold increases monotonically with R_{pp}, with a growth rate change at the inflection point. A piecewise linear threshold T function is accordingly constructed as follows:

$$T\left(R_{pp}\right) = \begin{cases} 0.2 + 0.4 \cdot \left(R_{pp} - 0.45\right), & R_{pp} < 0.45 \\ 0.2 + 0.85 \cdot \left(R_{pp} - 0.45\right), & R_{pp} \geq 0.45. \end{cases} \tag{17}$$

Under this category, an echo block requires full imaging when $R_{vp} < T(R_{pp})$.

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The second criterion applies to single-peak histograms, where the above bimodal criteria no longer apply. Let P_{num} denote the number of dominant peaks. As target energy merges with clutter and cannot be reliably discriminated in the raw echo domain, full imaging is performed when $P_{num} = 1$ for subsequent detection.

The third criterion triggers when the normalized position of the main peak along the amplitude axis X_{main} exceeds threshold $T_3 = 0.3$. This covers dense open-ocean ship clusters and nearshore strong-scattering zones, where basic echo-domain judgment carries high missed-detection risk. Full imaging is therefore mandatory, followed by precise detection and localization via the detection module, to balance efficiency and reliability.

Collectively, the final decision rule for full imaging processing is a logical OR combination of the three criteria above. An echo block will be subject to the full imaging pipeline if any of the following conditions is satisfied:

$$\mathbb{I}_{image} = \mathbb{I}\left(R_{vp} < T\left(R_{pp}\right)\right) \cup \mathbb{I}(P_{num} = 1) \cup \mathbb{I}(X_{main} < T_3) \quad (18)$$

where $\mathbb{I}_{image}$ denotes the decision result for imaging necessity, $\mathbb{I}_{image} = 1$ indicates that imaging is required for this region, and otherwise, imaging is not performed.

This target existence prediction mechanism effectively reduces the likelihood of target-free azimuth blocks undergoing full imaging processing and fundamentally cuts down the overall computational load of the integrated imaging-detection pipeline.

B. Interblock Parallel Pipeline With Intrablock Serial Execution

Building on the echo-domain prejudgment criteria above, this section details the proposed integrated imaging-detection architecture, which adopts azimuth-block partitioning to enable independent parallel processing and significantly boost overall throughput. The intrablock processing flow and system-wide cross-block scheduling logic are presented in sequence.

Constrained by the inherent data dependency of SAR signal processing, a serial judgment pipeline is adopted within a single azimuth block: echo-domain target existence prediction is first performed on raw echo data, and subsequent imaging and target detection are executed sequentially only if potential targets are detected. The detailed workflow of this pipeline is illustrated in Fig. 7.

To handle interblock processing across azimuth blocks, this work designs a pipeline-based parallel thread architecture with three sequential tasks: echo-domain imaging prescreening, SAR imaging, and ship target detection.

Benefiting from the continuous and uniform echo acquisition of stripmap-mode spaceborne observation, each azimuth block serves as an independent task unit fed into the pipeline. The framework consists of three core threads: the preimaging judgment thread, the imaging thread, and the detection thread. This "block-as-task" paradigm is highly compatible with the pipeline scheduling mechanism. The detailed multithread parallel scheduling workflow is illustrated in Fig. 8, demonstrated with on-orbit satellite data.

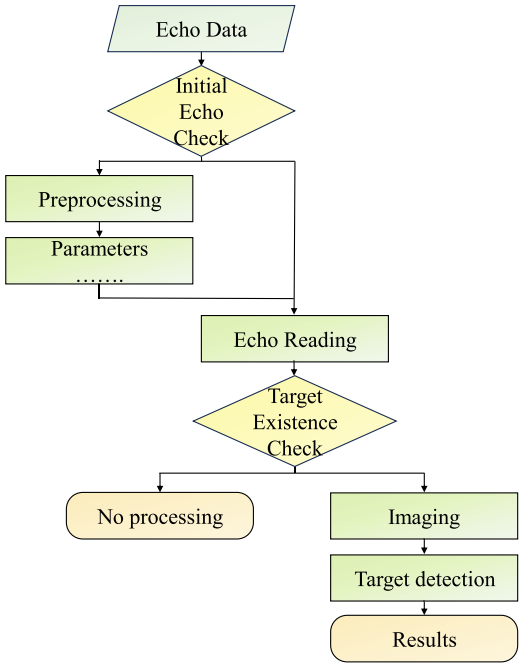


Fig. 7. Decision process diagram.

As shown in the figure, the echo-domain target screening is not perfectly accurate and may introduce false alarms, thus requiring secondary verification via postimaging detection.

This block-based parallel architecture, combined with the echo-domain prescreening mechanism, fundamentally improves the overall cascaded processing efficiency of the integrated imaging-detection system.

IV. Experiments and Results

A. Stage-Wise Validation Results

This section presents the stage-wise validation results of the proposed integrated imaging-detection architecture, with the core objective of independently verifying the functional correctness and precision performance of each key module. The validation covers three core components sequentially: the target existence prediction mechanism, the no-RCMC simplified imaging module, and the pruned ship detection network.

1) Target Existence Prediction Reliability Validation: To quantitatively evaluate the discrimination accuracy of the proposed target existence prediction mechanism, systematic experiments are performed to verify that it can effectively filter out target-free azimuth blocks while ensuring no missed detection for medium-to-large targets. Real spaceborne SAR echo data covering multiple typical scenarios are used to validate the proposed target existence prediction mechanism.

A sliding window strategy is adopted for azimuthal segmentation of the raw echo data. The block length is determined in accordance with the azimuth partitioning principle presented above, with a typical value of 4096 pixels and a sliding stride of 1000 pixels between consecutive blocks. Each azimuth block is first fully imaged via the standard CS algorithm, and

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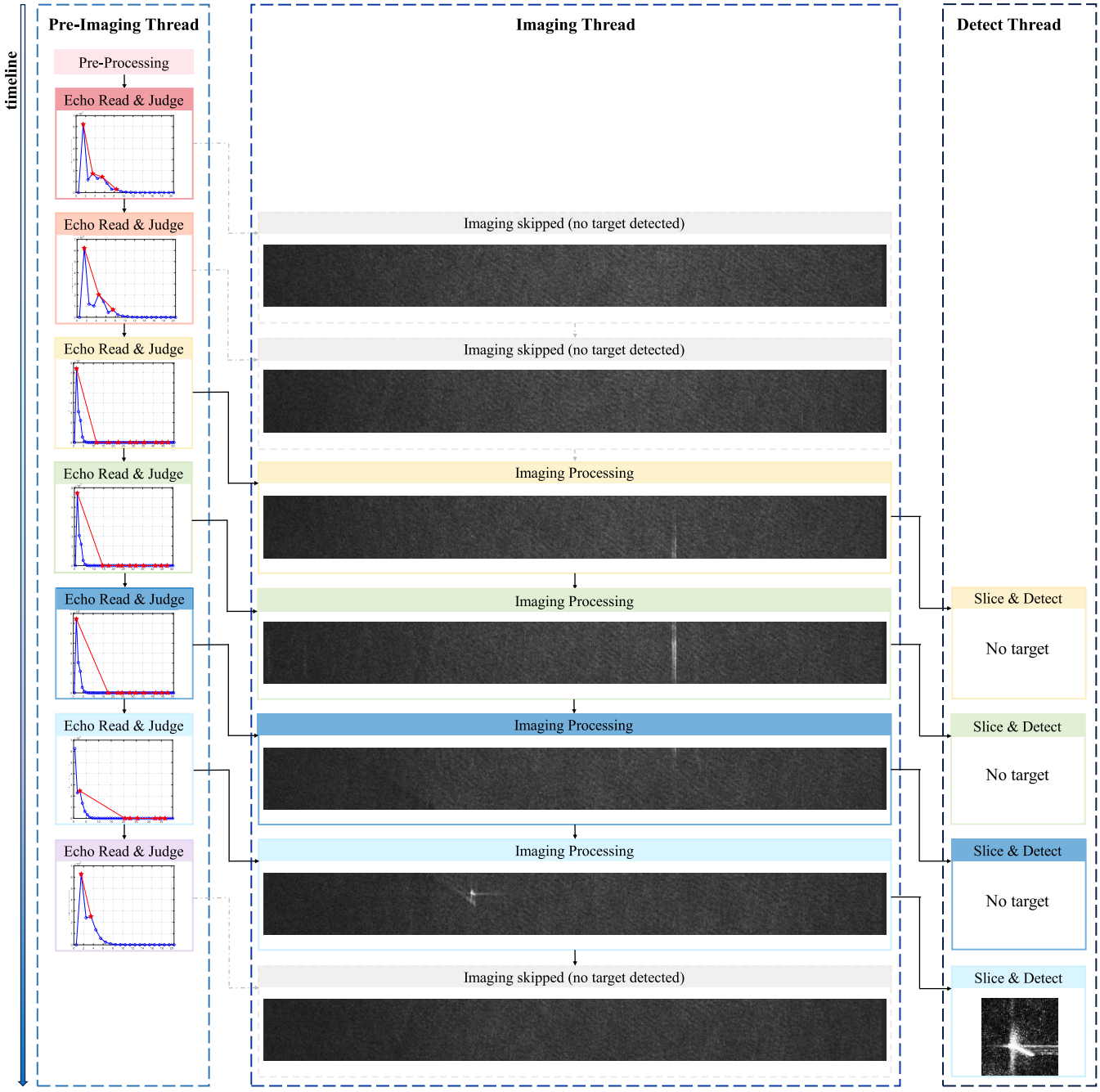


Fig. 8. Parallel processing flowchart of the scene-aware pipeline for integration of imaging detection. Task blocks in the same row are executed in parallel. Arrows indicate execution dependencies: a task starts only after all preceding tasks at the arrow origins are completed. Task blocks of the same color correspond to the same azimuth echo block, while gray blocks denote echo data judged to require no imaging, which bypass the full imaging pipeline. Specifically, the preimaging thread generates histograms for each azimuth strip after segmentation; the imaging thread outputs SAR images of echo blocks selected for full processing; the detection thread produces ship detection results from the corresponding imaged strips. The final imaging result in the figure is for demonstration only and does not represent actual operational output.

the imaging results are manually annotated to mark blocks containing ship targets or distinct coastal land-sea boundary scenes. Three characteristic metrics are extracted from the raw echo of each block: the valley depth between adjacent dominant peaks, the peak intensity ratio, and the main peak position. The optimal judgment thresholds are determined accordingly, and the final statistical classification results are summarized in Table II.

Metrics are defined as follows: “total echoes:” azimuth blocks generated by the aforementioned segmentation scheme across all scenarios; “imaged echoes:” blocks undergoing imaging processing; “target echoes:” target-containing blocks requiring imaging; “actual skipped ratio:” proportion of algorithm-judged imaging-exempt blocks relative to total echoes; “theoretical skipped ratio:” theoretical maximum skip rate with all target-free blocks skipped.

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TABLE II
STATISTICAL SUMMARY OF ECHO SELECTION FOR THE SAIDI METHOD

Scenes	Open_Sea	Near_Shore
Total Echoes	773	315
Imaged Echoes	631	234
Target Echoes	617	229
Actual Skipped Ratio(%)	18.37	25.71
Theoretical Skipped Ratio(%)	20.18	27.30
Recall	0.978	0.979
Precision	0.859	0.827
F1	0.915	0.896

TABLE III
POINT TARGET ANALYSIS METRICS OF TWO IMAGING ALGORITHMS

	CS		No-RCMC	
Dimension	Azimuth	Range	Azimuth	Range
Resolution	2.73	1.32	2.79	1.33
PSLR(dB)	-13.26	-13.28	-13.74	-13.51
ISLR(dB)	-10.01	-9.88	-10.79	-10.14

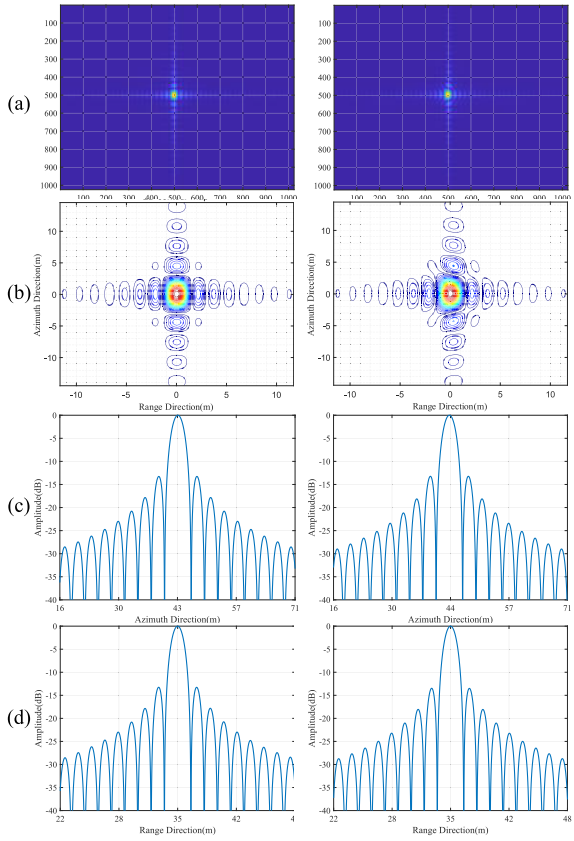


Fig. 9. Imaging results of point targets by the CS and No-RCMC algorithms. Columns 1 and 2 show the point target results obtained by the CS and No-RCMC algorithms. (a)–(d) Imaging map, contour map, azimuth profile, and range profile, respectively.

As shown in Table II, the proposed method achieves excellent recall performance in echo imaging necessity judgment, correctly identifying and processing nearly all target-bearing echoes that require imaging. The few missed targets are attributed to inherently weak target intensity, cloud cover, and strong sea clutter under rough sea conditions. Even so, the proposed partitioning scheme delivers favorable classification performance in the raw echo domain.

Overall, while ensuring virtually no missed detection of target-bearing echoes, this module achieves precompression of the overall computational load by embedding echo-domain target screening upstream of the full imaging pipeline, fully validating the effectiveness of the proposed design.

2) No-RCMC Imaging Accuracy Validation: To verify the detection feasibility of simplified No-RCMC imaging, this section quantifies quality degradation to ensure compatibility with subsequent detection. Since ship detection is the core application scenario, the imaging strategy in this work prioritizes detection efficiency while ensuring target detectability, rather than pursuing high image quality. For stripmap-mode data with limited resolution, omitting RCMC degrades image quality but does not significantly impair detection performance. The following experiments verify the feasibility of this simplified imaging scheme from two dimensions: quantitative evaluation via point target simulation, and qualitative verification via real measured data.

For the point target experiment, echo data are imaged by both the standard CS algorithm and the proposed No-RCMC algorithm after simulation. The focusing performance metrics of the two imaging results are calculated and compared, with detailed results shown in Fig. 9 and Table III.

For the real data experiment, 3-m resolution stripmap-mode data from on-orbit satellites are processed with the two algorithms above. Ship target slices are extracted from the imaging results for visual comparison, as presented in Fig. 10.

Qualitative comparison of both simulated point-target and real-measured scene results shows that omitting RCMC causes moderate image quality degradation, but the visual difference is negligible for image interpretation.

Accordingly, this simplified imaging strategy imposes no material adverse effect on target detection performance under open-sea stripmap-mode conditions. Overall, the proposed No-RCMC imaging scheme reduces the computational complexity of the imaging pipeline while maintaining target detectability of the generated images.

3) Pruning-Based Detection Accuracy Validation: To validate the pruned detection network, this section measures compression-induced accuracy loss to confirm it meets the required detection performance. To verify the performance of the pruned detection network, comparative experiments are conducted between the unpruned baseline network trained directly and the optimized network obtained via constrained training, channel pruning, and fine-tuning recovery. Detailed training configurations are listed in Table IV.

All models are initialized with public pretrained YOLOv11n weights and trained on the self-built FAST_SHIP dataset. Quantitative evaluation of both models is performed on the FAST_SHIP test set, with results summarized in Table V.

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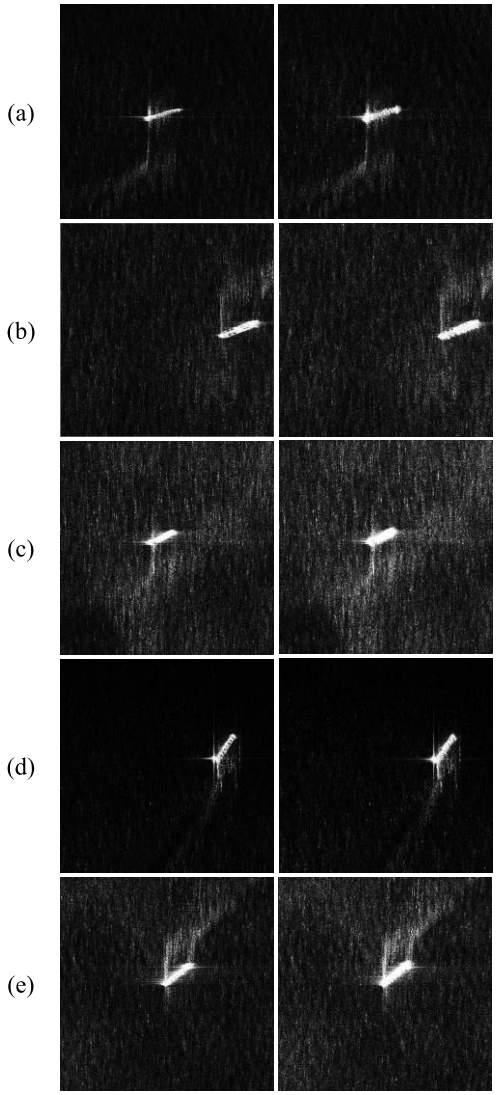


Fig. 10. Imaging results chart from a real satellite by the CS and No-RCMC algorithms. (a)–(e) are cropped patches of five distinct ship targets. Columns 1 and 2 show the ship target results obtained by the CS and No-RCMC algorithms.

TABLE IV
CONSTRAINED TRAINING PARAMETERS OF YOLOV11N ON THE FAST_SHIP DATASET

Parameters	Numerical Value
Optimizer	SGD
Initial Learning Rate	0.01
Batch Size	16
Baseline Epochs	192
Constraint Training Epochs	186
Channel Pruning Ratio	0.4
Fine-Tuning Epochs	34
Loss Function	CIoU + BCE (sigmoid) + DFL
Input Patch Size	1024*1024

Although the dataset scale is moderate and the scenario diversity is limited, which restricts validation sufficiency, and the model's generalization capability still requires further verification in more complex scenarios, the above results sufficiently demonstrate the effectiveness and feasibility of the proposed simplified imaging and detection scheme. This work also provides a reference for dataset construction in related research fields. Overall, the channel pruning operation compresses the computational load of the target detection module, laying a foundation for overall efficiency improvement, while the induced precision loss is limited and within an acceptable range.

TABLE V
VALIDATION SET EVALUATION RESULTS FOR TRAINED

	Pre-Pruning	Fine-Tuned
Image	290	290
Instance	293	293
Precision	0.937	0.868
Recall	0.898	0.895
mAP50	0.963	0.922
mAP50-95	0.615	0.539

Building on the three sections detailed above, Fig. 11(a) and (b) presents the schematics of the proposed SAIDI scene-aware processing pipeline for coastal and open-sea scenarios, respectively, to intuitively illustrate its operational workflow.

B. Results Evaluation

Building on the module-level validation presented earlier, this section conducts a systematic quantitative evaluation of the proposed SAIDI method from a cascaded processing perspective, focusing on temporal efficiency and hardware resource footprint for on-board real-time SAR deployment. To ensure the fairness and reproducibility of the comparative experiments, this section elaborates on the comparison baseline methods, the experimental platform and test conditions, as well as the definition and statistical criteria of each evaluation metric.

Experiments are performed on an NVIDIA Jetson Orin AGX with a 12-core Arm Cortex-A78AE CPU, 64 GB LPDDR5 memory, and a 1 TB NVMe SSD. The system runs Ubuntu 22.04 LTS (JetPack 6.2). Algorithms are implemented in C/C++ and OpenCV 4.8.0, compiled with GCC with -O3 optimization. A total of 59 scenes (18 near-shore, 41 open-sea) are tested under identical settings.

Three comparative schemes are designed for the cascaded processing performance benchmark to clarify the performance contribution of each optimization component. The traditional baseline scheme adopts the full CS imaging algorithm combined with the original unpruned detection network, serving as the performance reference without any structural simplification or target existence prediction mechanism. The simplified scheme uses the No-RCMC simplified imaging algorithm paired with the channel-pruned and fine-tuned detection network to evaluate the standalone effect of imaging simplification and network compression. The proposed SAIDI integrated framework further incorporates the echo-domain target existence prejudgment module and the pipeline framework, constructing the complete imaging-detection pipeline of this

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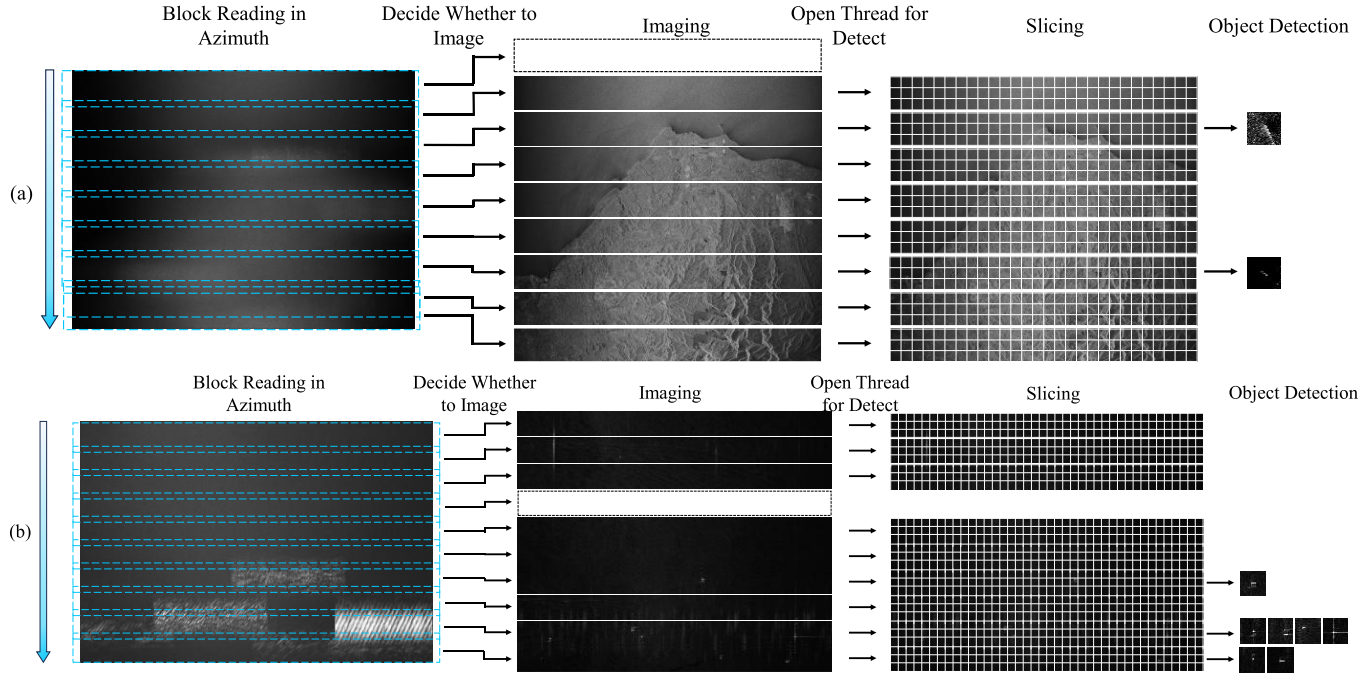


Fig. 11. Execution results of the scene-aware imaging-detection integrated pipeline for diverse scenarios. (a) Operation flowchart of scene-aware pipeline in near-shore scenario. (b) Operation flowchart of scene-aware pipeline in an open-sea scenario.

work. To ensure consistent deployment and acceleration conditions for all three schemes and eliminate irrelevant variables in performance comparison, the detection model weights of all schemes are uniformly exported in ONNX format and deployed for CPU-only inference in C/C++ via the open-source ONNX runtime, with a fixed batch size of 1 and a uniform input resolution of 1000 × 1000. CPU parallel optimization is applied to both the imaging and detection stages of the entire processing pipeline to improve overall processing efficiency.

The evaluation metrics cover two dimensions: temporal performance and hardware resource occupation. Processing latency is defined as the duration from raw echo input to final detection output, recorded via system-level timestamps. All hardware metrics are sampled at a fixed 500-ms interval throughout the entire processing workflow. CPU utilization is calculated by the following formula:

$$CPU\% = \frac{\frac{current_CPUtime - previous_CPUtime}{CLK_TCK}}{current_walltime - previous_walltime} \times 100\%. \quad (19)$$

This metric reflects the exploitation efficiency of CPU computing resources. Furthermore, the high-load duration ratio is defined as the proportion of total runtime where CPU utilization exceeds 90% of its peak value, to characterize the sustained high-load operating level of the pipeline. Peak memory refers to the maximum resident set size of the target process during the full execution cycle from echo reception to detection completion, representing the actual physical memory footprint of the pipeline. For power consumption, a no-load baseline is derived by averaging 60 instantaneous power samples collected over 30 s before algorithm execution.

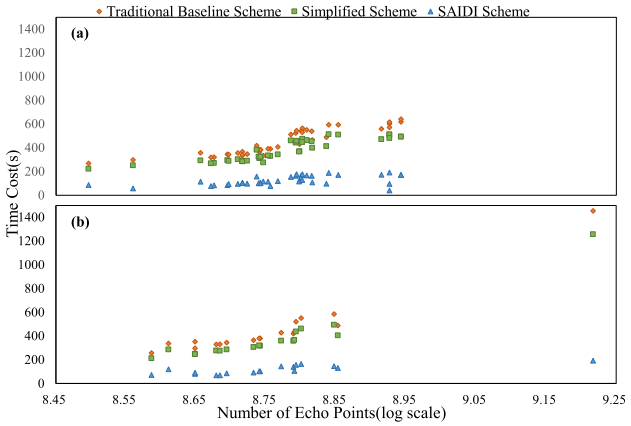


Fig. 12. Time costs of three comparative schemes. (a) Results for open-sea scenes. (b) Results for near-shore scenes.

Total runtime power is calculated via numerical integration of sampled data, and the energy saving ratio is computed against the traditional baseline scheme.

In terms of processing efficiency performance, the processing latency of the three comparative schemes is quantitatively analyzed to verify the efficiency improvement brought by each optimization component, with detailed results listed in Fig. 12.

Visualization results in Fig. 12 clearly demonstrate that the proposed SAIDI method holds a prominent processing efficiency advantage, with significantly lower time consumption than conventional imaging schemes. Specifically, its runtime accounts for 7.01%–37.36% of the traditional baseline scheme and 7.83%–42.74% of the simplified scheme, and the actual speedup varies with scene target sparsity.

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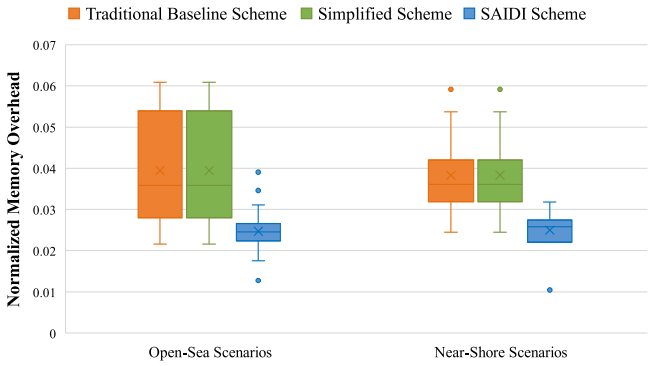


Fig. 13. Peak memory usage comparison.

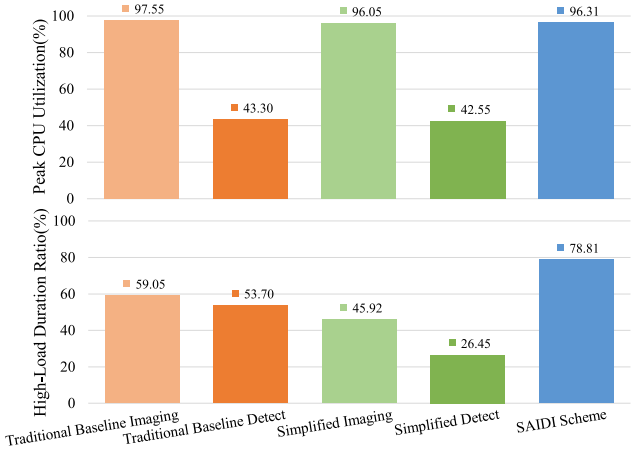


Fig. 14. CPU utilization comparison.

Building on the efficiency performance comparison, the hardware resource occupation of each scheme is further evaluated from three aspects to form a comprehensive assessment of on-board deployment adaptability.

Boxplots are adopted in Fig. 13; the ordinate represents the normalized memory overhead, that is, the peak memory(KB) per echo point during actual operation, which measures the memory cost per unit data scale and reflects the algorithm's resource utilization efficiency.

As shown in the figure, simplification of individual imaging or detection steps alone yields no significant reduction in peak memory. By contrast, the SAIDI scheme reduces peak memory to below 70% of the traditional baseline for 72% of samples. For a small fraction of samples (~15%) where no imaging steps are omitted, peak memory is comparable to or even slightly higher than that of the traditional method. Overall, the SAIDI method effectively reduces memory overhead in most scenarios while preserving full imaging integrity.

Fig. 14 evaluates peak CPU utilization and high-load duration ratio via two vertically stacked subplots.

The upper subplot shows that, given the limited number of on-board CPU cores, imaging methods with large data payloads readily drive CPU utilization to a high level. As a result, the traditional imaging method, the simplified No-RCMC imaging method, and the proposed SAIDI method all deliver

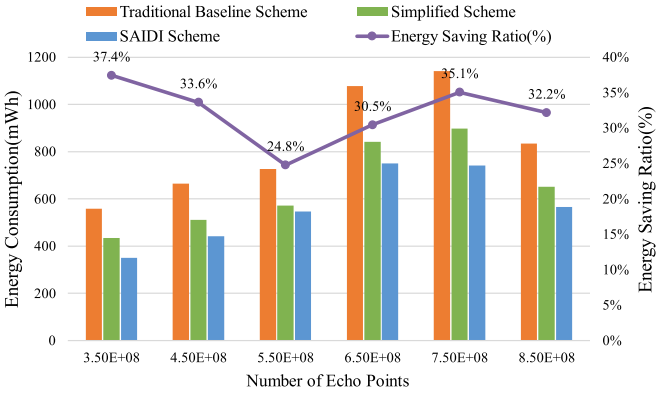


Fig. 15. Energy consumption and the energy saving ratio.

comparably high peak CPU utilization. The lower subplot compares the high-load duration ratio, which quantifies the sustained high-load operating level of each scheme. The comparison indicates that standalone simplification of individual imaging or detection modules leads to less efficient utilization of limited on-board computing resources. By contrast, the proposed method achieves superior performance in both peak CPU utilization and high-load duration ratio, enabling more adequate exploitation of constrained on-board resources.

Fig. 15 presents the energy consumption and energy saving ratio of the three methods, with data samples grouped by echo point count.

For 76.3% of the samples, the energy consumption of the SAIDI method drops below 80% of that of the traditional baseline. Additionally, for 72% of the samples, the energy consumption of the SAIDI method is lower than that of the simplified scheme. These findings demonstrate that the achieved energy savings stem from both the simplification of individual imaging and detection steps and the innovation of the overall processing architecture, effectively reducing power demand under limited on-board resource constraints.

The above results demonstrate the structural advantages of the proposed SAIDI integrated framework: it significantly reduces processing latency while maintaining controllable hardware resource consumption, well matching the tight resource constraints of on-board real-time SAR processing.

Overall, the experiments in this section systematically validate performance from module-level functional accuracy to system-level comprehensive metrics. The results consistently confirm that the proposed integrated imaging-detection scheme effectively balances processing efficiency, detection performance, and hardware overhead, with strong engineering feasibility for on-board real-time ship detection applications.

V. Conclusion

This study set out to present SAIDI, a novel integrated cascaded processing framework that unifies SAR imaging and ship detection to circumvent the latency of traditional workflows. The core innovation lies in a histogram-based echo screening mechanism, which selectively triggers imaging only for segments containing potential targets, thereby eliminating redundant computations. To further enhance efficiency,

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we omitted range migration correction and implemented a lightweight, pruned YOLO backbone coupled with a multi-thread parallel scheduling strategy. Additionally, an azimuthal block-based processing strategy was introduced to minimize memory footprint and output latency.

Experimental results validate the superiority of the proposed method. In different scenarios, SAIDI successfully bypasses target-free regions, reducing the overall runtime to 7.01%–37.36% of the traditional baseline scheme while cutting power consumption by over 24.8%. Notably, SAIDI enables more adequate exploitation of constrained on-board resources, and peak memory demand is reduced in 88.14% of test cases.

Despite these advancements, certain limitations remain. The screening metric needs broader validation across diverse platforms, and the conservative screening strategy retains minor computational redundancy. Furthermore, the current numerical thresholds are calibrated specifically for 3-m-resolution spaceborne offshore scenes. To enable cross-scene adaptability of the proposed method, a dedicated recalibration of these thresholds should be conducted for SAR data acquired by different sensors and at varying resolutions, taking into account the corresponding system parameters and sea clutter characteristics. Besides, the proposed method may suffer from limited sensitivity to isolated small ships, which could be missed during echo-domain prescreening or subsequent detection.

Future work will prioritize three strategic directions: integrating deep learning to enhance the adaptability of imaging necessity assessment, leveraging heterogeneous computing to maximize system throughput, and ultimately validating the framework on platforms for real-world on-orbit deployment.

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